

Manufacturing Engineering Technology I

TEKS §127.813 (working draft) · CTE Level 2 · Grade 10 · Mr. Gavaldon · Da Vinci School

Week 1**MONDAY****Day 1 - Procedures, Lab Safety & Design Notebook**

Aug 10

OBJECTIVE

Students will classify manufacturing-lab hazards, apply the NIOSH hierarchy of controls to select the most effective control for a given task, complete a Job Safety Analysis (JSA) with a likelihood x severity risk score, and set up a dated, numbered engineering design notebook as a professional record.

TEKS

§127.813(c)(1) safety, professional standards & responsible tool use

ELPS / EB SUPPORT

ELPS/EB supports at grade level: hazard-class icons and a live tool demo (VC, DP); pre-teach hierarchy-of-controls terms with a labeled pyramid before use (PV); JSA drafted with a bilingual peer and sentence stems for the 'control + justification' cell (PN, CD); wait time and rephrase/slow on the risk-scoring steps (WT, RS); extra time to finish the notebook set-up (ET).

AGENDA • 45 MIN

5' Bell-work: list every hazard you can spot in a shop photo

10' Why safety is a professional standard + the six hazard classes

12' Hierarchy of controls, machine guarding, LOTO & PPE

10' Worked Job Safety Analysis + guided you-try

5' The engineering notebook as a legal/professional record — set up Year-2 sections

3' Exit ticket + sign safety agreement

SLIDES / ACTIVITY

Working Safely: Hazards, Controls & the Professional Record

NOTEBOOK

Set up cover, table of contents, numbered/dated pages. First entry: one thing you want to design and build this year.

TURN IN

Job Safety Analysis (JSA) + Risk Scoring, Lab-Safety Agreement & Non-Negotiables

OBJECTIVE

Students will model production as an input -> process -> output system with a control loop, identify the six inputs for a real product, classify job/batch/repetitive/continuous production against the volume-variety product-process matrix, and compute a value-added ratio and per-unit cost to justify a production-type choice.

TEKS

\$127.813 (draft, verify strand) manufacturing systems - inputs, processes, outputs & production types

ELPS / EB SUPPORT

ELPS/EB supports at grade level: illustrated systems diagram and a picture word-bank for the six inputs (VC, DP, PV); the video gets an active-viewing note-catcher with sentence stems (CD); classification and the VA-ratio steps done in pairs so EB students reason aloud (PN); slow/rephrase the economies-of-scale graph and allow extra time on the calculation (RS, WT, ET).

AGENDA • 45 MIN

5' Bell-work: how is making a car different from a wedding cake?

8' The systems model + the six inputs + process families

7' Value-added vs the 7 wastes + worked VA-ratio

8' Toyota Production System video + active-viewing notes

7' Production types, product-process matrix & economies of scale

7' Guided practice: classify & justify + start worksheet

3' Exit ticket

SLIDES / ACTIVITY

Manufacturing Systems: Inputs, Processes & Production Types

NOTEBOOK

Draw the input -> process -> output diagram. List one product for each production type (job, batch, continuous).

TURN IN

Production Systems Analysis: Inputs, Value-Added & Cost

OBJECTIVE

Students will compute station cycle times, identify the bottleneck that caps a line's throughput, calculate throughput (units/hr and per shift) and takt time from customer demand, decide whether a line can meet demand, and propose a line-balancing change and recompute the new throughput.

TEKS

\$127.813 (draft, verify strand) production planning, logistics & workflow

ELPS / EB SUPPORT

ELPS/EB supports at grade level: color-coded station diagram with the bottleneck highlighted (VC, DP); pre-teach cycle time / takt / throughput with a worked model before independent work (PV); worksheet calculations started with a bilingual partner and a formula reference card (PN, CD); wait time on the two-step feasibility comparison and extra time on the recompute (WT, ET, RS).

AGENDA • 45 MIN

5' Bell-work: define a bottleneck in your own words

10' Cycle time, bottleneck & throughput + worked example

8' Takt time & feasibility + guided you-try

7' Line balancing, lead time, WIP & JIT

12' Production-Line Analysis worksheet

3' Exit ticket

SLIDES / ACTIVITY

Planning & Logistics: Cycle Time, Bottlenecks & Takt

NOTEBOOK

Draw a workflow diagram for a simple product (e.g., a sandwich shop). Label one bottleneck.

TURN IN

Production-Line Analysis: Bottleneck, Throughput & Takt

OBJECTIVE

Students will describe Industry 4.0 and the El Paso-Juarez manufacturing region, compare manufacturing career pathways using approximate wage and education data, calculate a simple return-on-investment/payback on a training credential, and interpret an SDS, GHS pictograms, and an NFPA 704 diamond to select required PPE and a safe handling decision.

TEKS

\$127.813 (draft, verify strand) industry overview, careers & workplace safety

ELPS / EB SUPPORT

ELPS/EB supports at grade level: hazard pictograms and the NFPA color diamond taught visually with a legend (VC, DP, PV); the factory video gets an active-viewing note-catcher (CD); career research uses a template with sentence stems and a bilingual partner (CD, PN); the ROI/payback calculation modeled first, with a formula card and extra time (RS, ET); wage terms rephrased and slowed (RS, WT).

AGENDA • 45 MIN

5' Bell-work: name a product made in the El Paso-Juarez region

10' Industry 4.0, sectors & the borderplex + factory video

8' Career pathways, wages & education ROI

9' Hazard communication: SDS, GHS pictograms & NFPA 704 + worked read

10' SDS analysis task + career analysis

3' Exit ticket

SLIDES / ACTIVITY

The Industry, Careers & Hazard Communication

NOTEBOOK

Note one manufacturing career: what they do, pay range, and education needed.

TURN IN

Hazard Communication: SDS, GHS & NFPA 704 Analysis, Manufacturing Career Analysis + ROI

OBJECTIVE

Students will apply the engineering design process to write a formal problem statement with measurable success criteria and constraints, generate three concepts, select a concept using a weighted decision matrix with computed weighted scores, apply Design-for-Manufacturing and tolerance thinking, and justify the chosen concept with evidence.

TEKS

\$127.813 (draft, verify strand) engineering design process applied to a manufacturable product

ELPS / EB SUPPORT

ELPS/EB supports at grade level: the design-process loop shown as a labeled diagram and the decision matrix modeled cell-by-cell (VC, DP); pre-teach constraint / criteria / tolerance / trade-off before use (PV); problem statement and matrix built with a bilingual partner using sentence stems, and drawings accepted for concepts (PN, CD, VC); extra time and a worked template for the weighted-score arithmetic (ET, RS).

AGENDA • 45 MIN

5' Bell-work: a product that annoys you - what's the real problem?

10' Engineering design process, constraints vs criteria & DFM

7' Tolerance + weighted decision matrix (worked)

20' Design brief: problem statement, concepts & decision matrix

3' Exit ticket + share

SLIDES / ACTIVITY

The Design Process: From Problem to a Manufacturable Concept

NOTEBOOK

Glue in the product-idea sheet. Sketch your top idea and list what it's made of.

TURN IN

Product Design Brief + Weighted Decision Matrix